

Homework6

1. Solve:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0,$$

$$u|_{x=0} = u_0, \quad u|_{x=a} = u_0 \left[3 \left(\frac{y}{b} \right)^2 - 2 \left(\frac{y}{b} \right)^3 \right],$$

$$\frac{\partial u}{\partial y} \Big|_{y=0} = 0,$$

$$\frac{\partial u}{\partial y} \Big|_{y=b} = 0,$$

(b=constant)

2. Solve:

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = bx(l-x),$$

$$u|_{x=0} = 0, \quad u|_{x=l} = 0,$$

$$u|_{t=0} = 0, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = 0.$$

3. Solve in rectangular region : $0 \leq x \leq a, -b/2 \leq y \leq b/2$

$$(1) \begin{cases} \nabla^2 u = -2, \\ u|_{x=0,a} = 0, \\ u|_{y=\pm b/2} = 0; \end{cases}$$

$$(2) \begin{cases} \nabla^2 u = -x^2 y, \\ u|_{x=0,a} = 0, \\ u|_{y=\pm b/2} = 0. \end{cases}$$

4. A slender beam is fixed at $x=0$, while the end at $x=l$ is subjected to a periodic force $A \sin \omega t$. Assuming the initial displacement and initial velocity are both zero, solve the transverse vibration problem of the beam.

5. Try to solve the following solution problem

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0,$$

$$u|_{x=0} = \cos \frac{\pi}{l} at, \quad \frac{\partial u}{\partial x} \Big|_{x=l} = 0,$$

$$u|_{t=0} = \cos \frac{\pi}{l} x, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = \sin \frac{\pi}{2l} x.$$

6. Try to solve the following solution problem

$$\frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0,$$

$$u|_{x=0} = A e^{i\omega t}, \quad u|_{x=l} = 0,$$

$$u|_{t=0} = 0.$$

7. Try to solve the following solution problem

$$\frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0,$$

$$u|_{x=0} = A \exp \{ -\alpha^2 \kappa t \}, \quad u|_{x=l} = B \exp \{ -\beta^2 \kappa t \},$$

$$u|_{t=0} = 0.$$